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Concomitant administration of a fully liquid, ready-to-use DTaP-IPV-HB-PRP-T hexavalent vaccine with a meningococcal serogroup C conjugate vaccine in infants

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ABSTRACT

DTaP-IPV-HB-PRP-T or hexavalent vaccines are indicated for primary and booster vaccination of infants and toddlers against diphtheria, tetanus, pertussis, hepatitis B, poliomyelitis and invasive diseases caused by *Haemophilus influenzae* type b (Hib). The present study evaluates the safety and immunogenicity of a ready-to-use hexavalent vaccine when co-administered with a meningococcal serogroup C conjugate (MenC) vaccine in infants.

This was a phase III, open-label, randomised, multicentre study conducted in Finland. Healthy infants, aged 46–74 days (n = 350), were randomised in a ratio of 1:1 to receive DTaP-IPV-HB-PRP-T vaccine at two, three and four months, either with a MenC vaccine co-administered at two and four months (Group 1; n = 175) or without MenC vaccine (Group 2; n = 175). All infants also received routine rotavirus and 13-valent pneumococcal conjugate vaccines.

The proportion of participants with an anti-HBs concentration ⩾10 mIU/mL assessed one month after the third dose of DTaP-IPV-HB-PRP-T vaccine was 97.5% [95%CI: 93.1–99.3] in the coadministration group and 96.1% [95%CI: 91.8–98.6] in the group without MenC vaccine. The proportion of participants with an anti-MenC SBA titre ⩾8 assessed one month after the second dose of MenC vaccine was 100% in the coadministration group. Both primary objectives were achieved.

Secondary immunogenicity and safety analyses showed that co-administration of DTaP-IPV-HB-PRP-T and MenC vaccines did not impact the immune response to the antigens of each of the two vaccines. All vaccines were well tolerated and the safety profile of DTaP-IPV-HB-PRP-T vaccine was similar in both groups.

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1. Introduction

Since the introduction of meningococcal serogroup C conjugate (MenC) vaccines in the national immunisation programmes for infants and toddlers, there has been a substantial decline in invasive meningococcal disease (IMD) caused by serogroup C [1–3]. In some European countries, immunisation against serogroup C related IMD is recommended with a 1-dose or 2-dose schedule administered concomitantly with DTaP-based combination vaccines during primary administration in the first year of life [4].

The number of immunisations recommended for children in the first two years of life has significantly increased in recent years [4]. Combination vaccines reduce the number of injections, simplify scheduling and may reduce anxiety and pain in children and increase acceptance among providers and parents. Overall, combination vaccines act to reduce further the burden of vaccine-preventable diseases through increased vaccine coverage rates [5–7].

DTaP-IPV-HB-PRP-T (Hexyon®, Hexacima®, Hexaxim®) is a fully liquid, ready-to-use hexavalent combination vaccine that received

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marketing authorisation in Europe in April 2013 and is indicated for primary and booster vaccination of infants and toddlers against diphtheria, tetanus, pertussis, hepatitis B (Hep B), poliomyelitis and invasive diseases caused by *Haemophilus influenzae* type b (Hib). DTaP-IPV-HB-PRP-T can be administered concomitantly with other paediatric vaccines given routinely at the same ages (rotavirus vaccine; pneumococcal conjugate vaccine; measles, mumps, rubella vaccine) [8]. The co-administration of this new hexavalent vaccine with meningococcal conjugate vaccine, however, has not yet been studied. The purpose of this study was to provide data on the co-administration of DTaP-IPV-HB-PRP-T and MenC vaccine. The study was designed to assess the immunogenicity and safety of DTaP-IPV-HB-PRP-T when administered concomitantly with a MenC vaccine during the first year of life, and to assess whether concomitant administration has any impact on the immunogenicity and safety of either vaccine.

2. Materials and methods

2.1. Design

This phase III, open-label, randomised, comparative, two-arm, multicentre study (clinicaltrial.gov: NCT01839175; EudraCT: 2012-005547-24) was conducted at 11 vaccine research clinics in Finland.

The study had two primary objectives. The first was to demonstrate that concomitant administration of DTaP-IPV-HB-PRP-T vaccine at two, three and four months of age with a MenC vaccine at two and four months of age (Group 1) was non-inferior to the administration of DTaP-IPV-HB-PRP-T vaccine without a MenC vaccine (Group 2) in terms of seroprotection rate for hepatitis B (defined as the proportion of participants with an anti-HBs concentration $\geqslant$10 mIU/mL) one month after the third dose of DTaP-IPV-HB-PRP-T vaccine. The second primary objective was to demonstrate that the concomitant administration of a MenC vaccine at two and four months of age with DTaP-IPV-HB-PRP-T vaccine at two, three and four months of age induced an acceptable response for MenC in terms of seroprotection rate (defined as the proportion of participants with MenC serum bactericidal antibody (SBA) titres $\geqslant$8) one month after a second dose of MenC vaccine.

The secondary immunogenicity objectives were to describe in each group the antibody response to all the hexavalent vaccine antigens one month after completion of the 3-dose primary schedule and in Group 1, the antibody responses to MenC vaccine one month after each dose of MenC vaccine.

Participants were randomised (1:1 ratio) to receive either DTaP-IPV-HB-PRP-T vaccine concomitantly with MenC vaccine (Group 1), or DTaP-IPV-HB-PRP-T vaccine alone (Group 2). A double-blind design, which would have required a placebo injection at Visit 1 and Visit 3 in Group 2, was deemed unnecessary as the primary endpoints were based on immunological criteria.

The study was conducted in accordance with applicable local and national requirements regarding ethical committee review, the International Conference on Harmonisation, Good Clinical Practice standards, the World Medical Association's Ethical Principles for Medical Research Involving Human Participants and the Declaration of Helsinki. Written informed consent was obtained from the parent(s) or legal guardian(s) of the participating children.

2.2. Study population

Healthy infants born at full term ($\geqslant$37 weeks) and/or with a birth weight $\geqslant$2.5 kg and aged $\geqslant$46 days and $\leqslant$74 days were eligible for enrolment.

The main exclusion criteria were the following: previous vaccination against diphtheria, tetanus, pertussis, Hep B, poliomyelitis, Hib diseases, meningococcal, pneumococcal and rotavirus infections; previous (within four weeks), or planned vaccination during study participation, any history of diphtheria, tetanus, pertussis, poliomyelitis, hepatitis B, Hib or MenC infections, or known personal or maternal history of hepatitis B (HBsAg) or hepatitis C seropositivity; known or suspected contraindication(s) to any of the study vaccines, hypersensitivity or allergy to the study vaccines' components; receipt of high doses of systemic corticosteroid therapy or other immunosuppressive therapy since birth; or any recent history of chronic disease or medical condition likely to interfere with the trial assessments.

2.3. Vaccines and vaccinations

DTaP-IPV-HB-PRP-T vaccine (Hexyon®, Sanofi Pasteur MSD, Lyon, France; Hexacima®, Hexaxim®, Sanofi Pasteur; Lyon, France; batch number: S4370) was supplied in pre-filled syringes. Each dose (0.5 mL) contained $\geqslant$20 IU diphtheria toxoid; $\geqslant$40 IU tetanus toxoid; 25 μg pertussis toxoid (PT); 25 μg filamentous haemagglutinin (FHA); 40, 8 and 32 D-antigen units of inactivated poliovirus vaccine (IPV) type 1, 2 and 3, respectively; 12 μg of Hib polysaccharide conjugated to tetanus toxoid; 10 μg Hep B surface antigen (HBsAg); and 0.6 mg aluminium hydroxide.

Each dose (0.5 mL) of MenC vaccine (NeisVac-C®, Baxter A.G.; batch number: VNSM04C) contained: 10 μg *Neisseria meningitidis* serogroup C polysaccharide conjugated to 10–20 μg tetanus toxoid; and 0.5 mg aluminium hydroxide and was supplied in pre-filled syringes.

Pneumococcal conjugate vaccine (Prevenar 13®, Pfizer; batch number: G40914) was supplied in pre-filled syringes (0.5 mL) and administered at two and four months of age, and rotavirus vaccine (RotaTeq®, MSD; batch number: H017807) was supplied in a pre-filled squeezable tube for oral administration at two, three and four months of age.

The DTaP-IPV-HB-PRP-T vaccine was administered intramuscularly in the anterolateral aspect of the left upper thigh; NeisVac-C® and Prevenar 13® were administered intramuscularly 5 cm apart in the right upper thigh. All vaccines were administered according to their respective summaries of product characteristics [8–11].

2.4. Immunogenicity measurements

At two months of age (prior to the first vaccination) a 3 mL blood sample was collected from all study participants for determination of anti-PT and anti-FHA antibody concentrations. A 3 mL sample was taken from Group 1 participants at three months of age (prior to further vaccination) for determination of MenC SBA titre one month after the first MenC vaccination. At five months of age a 5 mL sample was taken for assessment of antibodies to all DTaP-IPV-HB-PRP-T and MenC vaccine antigens.

Serological analyses for DTaP-IPV-HB- PRP-T vaccine antigens were performed by laboratory staff blinded to the group to which participants were randomised, at the Sanofi Pasteur Global Clinical Immunology Laboratory in Swiftwater, Pennsylvania, USA, or at qualified contract laboratories under Global Clinical Immunology supervision.

The commercially-available VITROS® hepatitis B enhanced chemiluminescence assay was used to measure HBsAg (anti-HBs) antibody concentrations. Anti-poliovirus types 1, 2, and 3 titres and anti-diphtheria antibody concentrations were measured by a micrometabolic inhibition test; anti-polyribosylribitol phosphate (PRP) antibody concentrations by radioimmunoassay. Enzyme-linked immunosorbent assay was used to measure anti-PT, anti-FHA and anti-tetanus antibody concentrations.

MenC antibodies were measured by a serum bactericidal antibody assay with baby rabbit complement (rSBA assay) at the Vaccine

Evaluation Unit, Public Health England, Manchester, UK. The antibody correlates of protection for each vaccine antigen were anti-HBs ⩾10 mIU/L, anti-PRP ⩾0.15 mg/mL, anti-diphtheria ⩾0.01 IU/mL, anti-tetanus ⩾0.01 IU/mL, anti-IPV 1, 2, 3 ⩾8, and anti-MenC SBA titre ⩾8.

2.5. Safety assessments

Investigators monitored safety and tolerability for unsolicited (spontaneously reported) systemic adverse events (AEs) occurring within 30 minutes after each vaccination. After each vaccination, participants' parents or legal guardians used a diary card to record solicited injection-site adverse reactions (injection-site erythema, injection-site pain, injection-site swelling; days 0–7), solicited systemic adverse reactions (pyrexia, vomiting, crying, somnolence, anorexia and irritability; days 0–7), and unsolicited injection-site reactions and systemic AEs (days 0–28). Body temperature (rectal) was measured daily. Parents or legal guardians recorded the start, end and intensity of any unsolicited events; the investigator assessed the relationship of these events to the vaccination. Serious AEs, including deaths, were recorded throughout the study.

2.6. Statistical methods

Non-inferiority in terms of seroprotection rate for HBs was to be demonstrated if the two-sided 95% confidence interval (CI; Wilson score method without continuity correction [12]) around the difference in seroprotection rate (Group 1 minus Group 2) excluded a decrease of 10% or more. The seroprotection rate to MenC was considered acceptable if the lower bound of its two-sided 95% CI (exact method for binary variables [13]) was >90%. Descriptive statistics were used for the secondary immunogenicity and safety analyses. The primary immunogenicity analysis was based on the per protocol set (PPS; all randomised participants excluding those with protocol violations that could interfere with the immunogenicity evaluation). Supportive analysis was provided on the Full Analysis Set (FAS; all randomised participants who received at least one dose of the study vaccine(s) and with any post-vaccination immunogenicity data). Safety analyses were described for all participants with safety follow-up data who received at least one dose of study vaccine(s) (SAS: safety analysis set). Statistical analyses were performed using SAS® software version 9.1.3 (SAS® Institute Inc., Cary, North Carolina, USA).

Sample size was calculated for the primary objectives of the study using the Farrington and Manning method [14] for the non-inferiority of anti-HBs seroprotection rates and based on the binomial distribution for the acceptability of MenC vaccine response rate. It was anticipated that up to 20% of the 350 planned participants would not be evaluable for the primary analysis (PPS), due to withdrawals and protocol deviations resulting in 140 evaluable participants per group (PPS). Study success required that the two primary objectives were achieved, therefore a type one error rate of 2.5% one sided was used. Based on expected response rates for hepatitis B non-inferiority and MenC acceptability gathered from previous studies with DTaP-IPV-HB-PRP-T [15] and MenC [16] (95% and 98%, respectively), and assuming independence of the primary hypotheses, the overall power for the success of the two primary objectives was approximately 90%.

3. Results

3.1. Study population

In total, 350 participants were enrolled (175 participants in each group). The study groups were comparable in terms of age, gender, ethnic origin, body weight and height. The mean age of the participants at vaccination visits 1, 2 and 3 was 63.0, 95.8 and 128.9 days, respectively, which confirmed a two, three and four-month vaccination schedule and was comparable between the groups.

The participant disposition is summarised in Fig. 1. The PPS comprised 322 participants (92.0%; Group 1, n = 162; Group 2, n = 160). Twenty-eight (8.0%) of the 350 randomised participants were excluded from the PPS: 13 (7.4%) in Group 1 and 15 (8.6%) in Group 2. Most of these individuals were non-compliant with their assigned schedule (n = 14; 4.0%: seven in each group) or blood sampling timetable (n = 14; 4.0%: six in Group 1 and eight in Group 2).

3.2. Vaccine immunogenicity

3.2.1. Immune response to the hexavalent vaccine antigens

3.2.1.1. Primary immunogenicity objectives. In the PPS, one month after dose 3 of DTaP-IPV-HB-PRP-T vaccine administered with or without MenC vaccine, the anti-HBs seroprotection rate was 97.5% and 96.1%, respectively. The non-inferiority of DTaP-IPV-HB-PRP-T vaccine administered concomitantly with MenC vaccine compared with DTaP-IPV-HB-PRP-T vaccine without MenC vaccine was demonstrated in terms of the seroprotection rate for HBs (proportion of participants with anti-HBs concentration ⩾10 mIU/mL) as the lower bound of the 95% CI of the difference (−2.92%) was greater than the pre-specified non-inferiority margin (−10%) (Table 1).

3.2.1.2. Secondary immunogenicity objectives. In the PPS, one month after dose 3 of DTaP-IPV-HB-PRP-T, no notable differences were observed between the two groups in seroprotection rates/response rates for any DTaP-IPV-HB-PRP-T antigen (Table 2). Proportions of participants showing antibody responses above the established correlates of protection were high and comparable for both groups (97.5–100% for Group 1 and 94.3–100% for Group 2). Similarly, anti-PT and anti-FHA vaccine responses were high and similar across groups (>99%). Approximately 90% of participants achieved a fourfold antibody rise for anti-PT and anti-FHA. Similarly, the GMTs or GMCs observed for hepatitis B, diphtheria, poliovirus and pertussis antigens were comparable for both groups although anti-PRP and anti-tetanus GMCs were numerically higher in Group1 (3.49 μg/mL: 95%CI [2.88; 4.24] and 1.17 IU/mL: 95%CI [1.07;1.29], respectively, than in Group 2 (1.89 μg/mL: 95%CI [1.49;2.38] and 0.78 IU/mL: 95%CI [0.70;0.87]), respectively (Table 2).

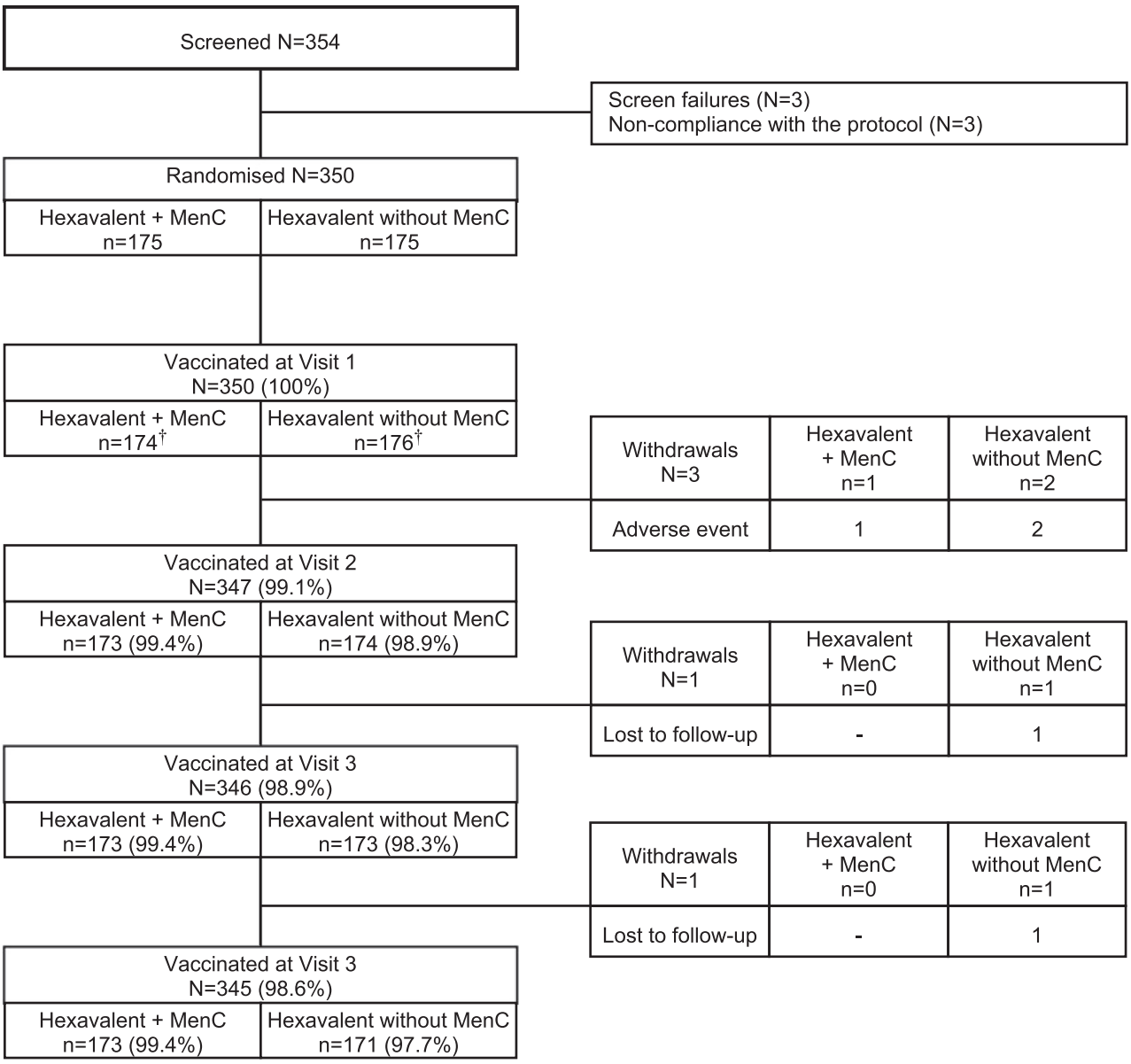
3.2.2. Immune response to the MenC vaccine

3.2.2.1. Immunogenicity primary objective. One month after dose 2 of MenC vaccine, the percentage of participants in the PPS with MenC SBA titre ⩾8 was 100% [95% CI: 97.7; 100.0], thus demonstrating acceptability of the seroprotection rate one month after dose 2 of MenC vaccine, as the lower bound of the 95% CI (97.7%) was greater than the pre-specified acceptability threshold (90%) (Table 1).

3.2.2.2. Immunogenicity secondary objective: post-dose 1 and post-dose 2 antibody response to MenC. The percentage of participants with MenC SBA titre ⩾8 was 99.4% one month after dose 1 of MenC and 100% one month after dose 2. The MenC SBA GMT was higher one month after dose 1 of MenC than one month after dose 2 with non-overlapping CI 95% (Table 3).

3.3. Safety results

All study participants reported at least one AE; the safety profile of DTaP-IPV-HB-PRP-T was similar in both groups (Table 4).



†One participant randomised to the Hexavalent + MenC Group was not vaccinated as per randomisation and did not receive MenC vaccination. As per safety set definition this subject was analysed according to the vaccine actually received and is therefore counted in the group hexavalent vaccine without MenC vaccine for the Safety Analysis Set.

Fig. 1. Disposition of study participants.

Table 1
Post-dose 3 anti-HBs seroprotection rate non-inferiority analysis and post-dose 2 anti-MenC seroprotection rate acceptability analysis (per protocol set).

	Group 1 DTaP-IPV-HB-PRP-T with MenC (N = 162)	Group 2 DTaP-IPV-HB-PRP-T without MenC (N = 160)	Seroprotection Group 1 − Group 2	Study outcome
Post-dose 3 anti-HBs seroprotection rate (n) ⩾10 mIU/mL	160	155		
n (%)	156 (97.5%)	149 (96.1%)	1.37%	Non-inferiority met[a]
[95% CI]	[93.7; 99.3]	[91.8; 98.6]	[−2.92; 5.95]	
Post-dose 2 MenC SBA seroprotection rate, per protocol set (n) ⩾8	162	n/a	n/a	
n (%)	162 (100%)	n/a	n/a	Acceptability met[b]
[95% CI]	[97.7; 100]			

Notes:
[a] Group 1 seroprotection rate is considered to be non-inferior to Group 2 seroprotection rate if the lower bound of the 95% CI of the difference is greater than −10%.
[b] Seroprotection rate is considered acceptable if the lower bound of the 95% CI is greater than the pre-specified acceptability threshold of 90%.

Table 2
Post-dose 3 antibody response and GMT/GMCs to all DTaP-IPV-HB-PRP-T antigens (per protocol set).

		Group 1 DTaP-IPV-HB-PRP-T with MenC (N = 162)			Group 2 DTaP-IPV-HB-PRP-T without MenC (N = 160)		
Component	Endpoint	n	Observed response	[95% CI]	n	Observed response	[95% CI]
Anti-HBs	Antibody response ⩾10 mIU/mL n (%)	160	156 (97.5%)	[93.7;99.3]	155	149 (96.1%)	[91.8;98.6]
	GMC (mIU/mL)	160	242.75	[195.05;302.11]	155	267.58	[212.65;336.70]
Anti-PRP	Antibody response ⩾0.15 μg/mL n (%)	160	157 (98.1%)	[94.6;99.6]	158	149 (94.3%)	[89.5;97.4]
	GMC (μg/mL)	160	3.49	[2.88;4.24]	158	1.89	[1.49;2.38]
Anti-diphtheria	Antibody response ⩾0.01 IU/mL	160	160 (100%)	[97.7;100.0]	158	157 (99.4%)	[96.5;100.0]
	Antibody response ⩾0.10 IU/mL	160	56 (35.0%)	[27.6;42.9]	158	65 (41.1%)	[33.4;49.2]
	GMC (IU/mL)	160	0.08	[0.07;0.10]	158	0.09	[0.08;0.10]
Anti-tetanus	Antibody response ⩾0.01 IU/mL	159	159 (100%)	[97.7;100.0]	156	156 (100%)	[97.7;100.0]
	Antibody response ⩾0.10 IU/mL	159	159 (100%)	[97.7;100.0]	156	155 (99.4%)	[96.5;100.0]
	GMC (IU/mL)	159	1.17	[1.07;1.29]	156	0.78	[0.70;0.87]
Anti-pertussis	Vaccine response[a]	154	152 (98.7%)	[95.4;99.8]	154	154 (100%)	[97.6;100.0]
	4-fold increase	154	136 (88.3%)	[82.2;92.9]	154	136 (88.3%)	[82.2;92.9]
	GMC (EU/mL)	160	129.74	[118.93;141.52]	159	139.91	[126.98;154.15]
Anti-FHA	Vaccine response[a]	154	153 (99.4%)	[96.4;100.0]	153	153 (100%)	[97.6;100.0]
	4-fold increase	154	138 (89.6%)	[83.7;93.9]	153	140 (91.5%)	[85.9;95.4]
	GMC (EU/mL)	158	123.54	[112.47;135.69]	156	147.77	[134.82;161.97]
Anti-IPV1	Antibody response ⩾8	159	159 (100%)	[97.7;100.0]	152	150 (98.7%)	[95.3;99.8]
	GMT	159	92.49	[75.06;113.97]	152	126.84	[101.46;158.56]
Anti-IPV2	Antibody response ⩾8	159	159 (100%)	[97.7;100.0]	152	152 (100%)	[97.6;100.0]
	GMT	159	90.9	[73.23;112.83]	152	104.72	[82.66;132.67]
anti-IPV3	Antibody response ⩾8	159	159 (100%)	[97.7;100.0]	152	151 (99.3%)	[96.4;100.0]
	GMT	159	173.3	[138.20;217.31]	152	250.78	[197.56;318.34]

Notes:
n: number of participants with data available.
[a] Pertussis vaccine response is defined as: – if pre-vaccination antibody concentration was <4xLLOQ, then the post-vaccination antibody concentration was to be ⩾4xLLOQ. – if pre-vaccination antibody concentration was ⩾4xLLOQ, then the post-vaccination antibody concentration was to be ⩾pre-immunisation levels.

Table 3
Post-dose 2 MenC antibody response rate and GMT (per protocol set).

		Group 1 DTaP-IPV-HB-PRP-T with MenC (N = 162)					
		Dose 1			Dose 2		
Antigen	Endpoint	n	Observed response	[95% CI]	n	Observed response	[95% CI]
MenC	SBA ⩾8	157	156 (99.4%)	[96.5;100.0]	162	162 (100%)	[97.7;100.0]
	SBA ⩾128	157	154 (98.1%)	[94.5;99.6]	162	156 (96.3%)	[92.1;98.6]
	SBA GMT	157	885.17	[737.06;1063.04]	162	579.64	[505.36;664.84]

Notes: n: number of participants with data available.

Solicited reactions occurring within 7 days after any vaccine injection were reported by all participants in both study groups. Within seven days after any vaccination, solicited injection-site adverse reactions were reported by 89.1% [95% CI: 83.5; 93.3] of participants in Group 1, and 73.9% [95% CI: 66.7; 80.2] of participants in Group 2. After DTaP-IPV-HB-PRP-T vaccination, the most frequent injection-site adverse reaction was injection-site pain (around 65%), followed by injection-site erythema (around 55%), and injection-site swelling (around 35%). No notable differences were observed between the groups with regard to the intensity of solicited injection-site adverse reactions.

All participants experienced at least one solicited systemic adverse reaction. Irritability was reported by most of the participants (95.4% in Group 1 versus 94.3% in Group 2). The frequency of crying and vomiting were the only noteworthy differences between the groups. Crying was reported more frequently in Group 1 than in Group 2 (85.1% versus 71.0%) with a similar trend in grade 3 intensity crying (>3 h). There was a higher frequency of vomiting in Group 1 than in Group 2 (36.2% versus 26.7%, respectively). Episodes of pyrexia of grade 3 intensity (>39.5 °C) were observed in two participants (1.1%) in Group 1.

Within 30 days of any vaccination, unsolicited adverse reactions were slightly more frequent in Group 1 (27.6%) than in Group 2 (22.7%) due to higher unsolicited non-serious adverse reactions in Group 1 than in Group 2 (injection-site induration: 8.0% versus 5.1%; diarrhoea: 6.3% versus 4.0%; injection-site bruising: 5.2% versus 4.0%, respectively). The incidence of unsolicited systemic reactions was similar in both groups (14.9% and 15.9%, respectively).

Six serious AEs were reported in 6 participants (1.7%): 4 participants (2.3%) in Group 1 and 2 participants (1.1%) in Group 2. Only one serious AE was considered related to the study vaccines (a fever ⩽39.5 °C occurring in a participant of Group 1 on the day of the first vaccination, lasting 6 h and leading to hospitalisation).

Overall, three participants withdrew from the study after the first vaccination due adverse events, including two non-serious AEs considered related to the study vaccines (asthenia of grade 3 intensity reported by one participant in Group 1 and crying of grade 3 intensity reported by one participant in Group 2) and one SAE considered not related to the study vaccines (skin neoplasm bleeding of grade 3 intensity reported by one participant in Group 1). No deaths were reported during the study.

4. Discussion

This randomised study evaluated the safety and immunogenicity of the concomitant administration of one hexavalent DTaP-IPV-HB-PRP-T vaccine at two, three and four months of age with one Men C vaccine at two and four months of age.

Table 4
Solicited injection-site reactions and solicited systemic reactions after any vaccine injection within 0–7 days (safety analysis set).

Participants with at least one:	Group 1 DTaP-IPV-HB-PRP-T with MenC (N = 174)		Group 2 DTaP-IPV-HB-PRP-T without MenC (N = 176)	
	n (%)	[95% CI]	n (%)	[95% CI]
Solicited injection site reaction	155 (89.1%)	[83.5;93.3]	130 (73.9%)	[66.7;80.2]
After DTaP-IPV-HB-PRP-T vaccine	147 (84.5%)	[78.2;89.5]	130 (73.9%)	[66.7;80.2]
Injection site erythema	95 (54.6%)	[46.9; 62.1]	98 (55.1%)	[47.4; 62.6]
Grade 3: ⩾50 mm	6 (3.4%)	[1.3; 7.4]	8 (4.5%)	[2.0; 8.8]
Injection site pain	120 (69.0%)	[61.5;75.7]	109 (61.9%)	[54.3; 69.1]
Grade 3: cries when injected limb is moved or the movement of the injected limb is reduced	16 (9.2%)	[5.3;14.5]	12 (6.8%)	[3.6; 11.6]
Injection site swelling	60 (34.5%)	[27.5;42.1]	61 (34.7%)	[27.7; 42.2]
Grade 3 ⩾50 mm	7 (4.0%)	[1.6;8.1]	9 (5.1%)	[2.4; 9.5]
After MenC vaccine	124 (71.3%)	[63.9;77.9]	–	–
Injection site erythema	78 (44.8%)	[37.3;52.5]	–	–
Grade 3 (as above)	1 (0.6%)	[<0.1;3.2]	–	–
Injection site pain	106 (60.9%)	[53.2; 68.2]	–	–
Grade 3 (as above)	8 (4.6%)	[2.0; 8.9]	–	–
Injection site swelling	45 (25.9%)	[19.5;33.0]	–	–
Grade 3 (as above)	3 (1.7%)	[0.4; 5.0]	–	–
Solicited systemic reaction	174 (100%)	[97.9; 100]	176 (100%)	[97.9; 100]
Crying	148 (85.1%)	[78.9; 90.0]	125 (71.0%)	[63.7; 77.6]
Grade 3: >3 h	11 (6.3%)	[3.2; 11.0]	4 (2.3%)	[0.6; 5.7]
Decreased appetite	98 (56.3%)	[48.6; 63.8]	96 (54.5%)	[46.9; 62.1]
Grade 3: refuses > 3 feeds or refuses most feeds/meals	3 (1.7%)	[0.4; 5.0]	5 (2.8%)	[0.9; 6.5]
Irritability	166 (95.4%)	[91.1; 98.0]	166 (94.3%)	[89.8; 97.2]
Grade 3: inconsolable	21 (12.1%)	[7.6; 17.9]	21 (11.9%)	[7.5; 17.7]
Pyrexia	126 (72.4%)	[65.1; 78.9]	127 (72.2%)	[64.9; 78.6]
Grade 3:>39.5 °C	2 (1.1%)	[0.1; 4.1]	0	–
Somnolence	144 (82.8%)	[76.3; 88.1]	151 (85.8%)	[79.7; 90.6]
Grade 3: sleeping most of the time or difficult to wake up	6 (3.4%)	[1.3; 7.4]	7 (4.0%)	[1.6; 8.0]
Vomiting	63 (36.2%)	[29.1; 43.8]	47 (26.7%)	[20.3; 33.9]
Grade 3: ⩾6 episodes per 24 h or requiring parenteral hydration	4 (2.3%)	[0.6; 5.8]	2 (1.1%)	[0.1; 4.0]

Notes:
n: number of participants experiencing the endpoint listed in the first column.
95% CI: the two-sided 95% CI of the percentage of participants presenting the considered event at least once is calculated using the exact binomial method.

The concomitant administration of the DTaP-IPV-HB-PRP-T vaccine with a MenC vaccine was highly immunogenic and did not impair the immune response to either vaccine. With respect to the anti-HBs seroprotection rate, administration of DTaP-IPV-HB-PRP-T vaccine concomitantly with MenC vaccine was non-inferior to the administration of DTaP-IPV-HB-PRP-T without a concomitant MenC vaccine one month after a 3-dose schedule. An acceptable response for MenC in terms of seroprotection rate was induced one month after the second dose of MenC vaccine in participants who received DTaP-IPV-HB-PRP-T with concomitant MenC vaccine.

One month after a 3-dose schedule, levels of seroprotection rates/response rates were high for any hexavalent vaccine antigen and similar between both groups. The seroprotection rates reported in the present study are consistent with those previously reported for this hexavalent vaccine administered alone (i.e. without pneumococcal conjugate vaccine and without MenC vaccine) at two, three and four months of age [15,17].

The GMTs and GMCs were also comparable between the groups for all the hexavalent vaccine antigens except for PRP and tetanus. The lower anti-PRP GMCs observed for Group 2 are consistent with data reported previously in infants who received DTaP-IPV-HB-PRP-T alone at the same 3-dose schedule (2.12 μg/mL 95%CI [1.62; 2.77]) [15]. Enhancement of the response to a Hib-tetanus toxoid conjugate vaccine by concomitant administration with a tetanus toxoid conjugate meningococcal vaccine have been also reported previously [16,18–20].

The higher anti-tetanus GMCs observed in Group 1 has also been reported previously when NeisVac-C® was co-administered with Infanrix hexa® (DTaP-IPV-HB/Hib vaccine) at two and four months of age [18] or Pediacel® (DTaP-IPV-Hib vaccine) at two, three and four months of age [16]. The higher GMCs observed in infants that received MenC-Tetanus Toxoid vaccine is explained by the tetanus toxoid component of the MenC conjugate, which provide an additional dose of tetanus antigen.

The immune response elicited by the MenC vaccine was high after the first and second doses and consistent with previous studies where MenC vaccine was administered with combined DTaP-IPV-HB/Hib vaccines [16,21,22]. One month after the first dose, 99.4% of the participants had MenC SBA titre ⩾8 and 98.1% had MenC SBA titre ⩾128. This is consistent with published data reporting a MenC SBA titre ⩾8 in 97% of participants receiving a single dose of NeisVac-C® at two months of age concomitantly with Pediacel® [19] and in >99% of participants receiving one dose of NeisVac-C® at four or six months of age concomitantly with Infanrix hexa® and Prevenar 13® [9].

In this study, the MenC SBA GMT was higher one month after dose 1 of MenC vaccine than one month after dose 2. The provision of immunogenicity data after one dose of MenC vaccine, is interesting for countries where a one-dose schedule is still recommended during the first year of life.

The safety profile of DTaP-IPV-HB-PRP-T was generally similar whether administered concomitantly with MenC vaccine or not, although there was a trend towards higher reactogenicity in the concomitant group. This finding may simply reflect the greater antigen load given to infants in this group.

5. Conclusions

In conclusion, these data support the administration of MenC vaccine either in a 1-dose or a 2-dose schedule when concomitantly administered with this new hexavalent DTaP-IPV-HB-PRP-T during primary immunisation.

Disclosures and contributions

Disclosures

Timo Vesikari was lead investigator supported by research grants, Ray Borrow performed the MenC assays supported by research grants. Xavier Da Costa is an employee of Sanofi Pasteur and may hold company stock and/or stock options. Florence Boisnard, Cécile Eymin, Stephen Lockhart and Patrick Richard were all Sanofi Pasteur MSD employees at the time the study was conducted.

Contributions

TV contributed to protocol review, participants enrolment, subjects data collection/acquisition, data interpretation. RB and XdC contributed to the laboratory analysis and data interpretation. FB, PR and SL contributed to the study design and development, data analysis and interpretation. CE contributed to the writing of the manuscript drafts. All authors critically reviewed and revised the manuscript drafts, approved the final version of the manuscript and take responsibility for the integrity of the data and accuracy of data analysis. All authors were involved in the decision to submit the manuscript to Vaccine for publication.

Role of the funding source

Sanofi Pasteur MSD provided financial support for the conduct of the research and preparation of the article and were involved in the study design, in the collection, analysis and interpretation of data and in the writing of the trial report.

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